

4 Part load conditions in cooling mode

4.1 General

For the purpose of calculation of application SEER and reference SEER / SEER_{on} as explained in Clause 6, the part load ratios mentioned below shall be based on the part load ratio formulas (1st column of Tables 2 to 5) and not on the rounded figures as mentioned in the 2nd column of these tables.

4.2 Air-to-air units

The part load conditions for determining the reference SEER (Equation (1)) and reference SEER_{on} (Equation (3)) are given in the following table:

**Table 2 — Part load conditions for reference SEER and reference SEER_{on}
calculation of air-to-air units**

	Part load ratio	Part load ratio %	Outdoor air dry bulb temperature °C	Indoor air dry bulb (wet bulb) temperatures °C
A	$(35-16)/(T_{designc} - 16)$	100	35	27(19)
B	$(30-16)/(T_{designc} - 16)$	74	30	27(19)
C	$(25-16)/(T_{designc} - 16)$	47	25	27(19)
D	$(20-16)/(T_{designc} - 16)$	21	20	27(19)

5 Part load conditions in heating mode

5.1 General

For the purpose of calculation of application SCOP and reference SCOP / SCOP_{on} / SCOP_{net} as explained in Clause 7, the part load ratios mentioned below should be based on the part load ratio formulas (1st column of Tables 6 to 35) and not on the rounded figures as mentioned in the 2nd column of these tables.

For the purpose of reference SCOP and reference SCOP_{on} / SCOP_{net}, there are three reference conditions: average (A), warmer (W) and colder (C).

The relevant Tdesignh values are defined as follows:

- Tdesign “average” dry bulb temperature conditions at -10 °C outdoor temperature and 20 °C indoor temperature;
- Tdesign “colder” dry bulb temperature conditions at -22 °C outdoor temperature and 20 °C indoor temperature ;
- Tdesign “warmer” dry bulb temperature conditions at +2 °C outdoor temperature and 20 °C indoor temperature,

and the relevant $T_{bivalent}$ is defined as follows:

- for the average heating season, the dry bulb bivalent temperature is $+2\text{ }^{\circ}\text{C}$ or lower;
- for the colder heating season, the dry bulb bivalent temperature is $-7\text{ }^{\circ}\text{C}$ or lower;
- for the warmer heating season, the dry bulb bivalent temperature is $+7\text{ }^{\circ}\text{C}$ or lower.

NOTE If the declared TOL is lower than the $T_{designh}$ of the considered climate, then it may be assumed that TOL is equal to $T_{designh}$.

5.2 Air-to-air units

The part load conditions for determining the reference SCOP (Equation (7)), reference $SCOP_{on}$ (Equation (9)) and reference $SCOP_{net}$ (Equation (10)) are given in the following table:

Table 6 — Part load conditions for reference SCOP, reference $SCOP_{on}$ and reference $SCOP_{net}$ calculation of air-to-air units for the reference heating season “A” = average

	A		Outdoor air dry bulb (wet bulb) temperatures $^{\circ}\text{C}$	Indoor air dry bulb temperature $^{\circ}\text{C}$
	Part load ratio	Part load ratio %		
A	$(-7-16)/(T_{designh} - 16)$	88	-7(-8)	20
B	$(+2-16)/(T_{designh} - 16)$	54	2(1)	20
C	$(+7-16)/(T_{designh} - 16)$	35	7(6)	20
D	$(+12-16)/(T_{designh} - 16)$	15	12(11)	20
E	$(TOL-16)/(T_{designh} - 16)$		TOL	20
F	$(T_{bivalent}-16)/(T_{designh} - 16)$		$T_{bivalent}$	20

Table 7 — Part load conditions for reference SCOP, reference $SCOP_{on}$ and reference $SCOP_{net}$ calculation of air-to-air units for the reference heating season “W” = warmer

	W		Outdoor air dry bulb (wet bulb) temperatures $^{\circ}\text{C}$	Indoor air dry bulb temperature $^{\circ}\text{C}$
	Part load ratio	Part load ratio %		
A	(not applicable)			
B	$(+2-16)/(T_{designh} - 16)$	100	2(1)	20
C	$(+7-16)/(T_{designh} - 16)$	64	7(6)	20
D	$(+12-16)/(T_{designh} - 16)$	29	12(11)	20
E	$(TOL-16)/(T_{designh} - 16)$		TOL	20
F	$(T_{bivalent}-16)/(T_{designh} - 16)$		$T_{bivalent}$	20

Table 8 — Part load conditions for reference SCOP, reference SCOP_{on} and reference SCOP_{net} calculation of air-to-air units for the reference heating season “C” = colder

	C		Outdoor air dry bulb (wet bulb) temperatures °C	Indoor air dry bulb temperature °C
	Part load ratio	Part load ratio %		
A	$(-7-16)/(T_{designh} - 16)$	61	-7(-8)	20
B	$(+2-16)/(T_{designh} - 16)$	37	2(1)	20
C	$(+7-16)/(T_{designh} - 16)$	24	7(6)	20
D	$(+12-16)/(T_{designh} - 16)$	11	12(11)	20
E	$(TOL-16)/(T_{designh} - 16)$		TOL	20
F	$(T_{bivalent}-16)/(T_{designh} - 16)$		Tbivalent	20

6 Calculation methods for reference SEER and reference SEERon

6.1 General formula for calculation of reference SEER

The calculation of the reference SEER that applies to all types of units is given by the following formula:

Reference SEER = reference annual cooling demand divided by the annual electricity consumption

This annual electricity consumption includes the power consumption during active mode, thermostat off mode, standby mode, off mode and that of the crankcase heater.

The power consumption during active mode is derived from the calculation of $SEER_{on}$. For determination of $SEER_{on}$, see 6.3

$$SEER = \frac{Q_C}{\frac{Q_C}{SEER_{on}} + H_{TO} \times P_{TO} + H_{SB} \times P_{SB} + H_{CK} \times P_{CK} + H_{OFF} \times P_{OFF}} \quad (1)$$

where

Q_C = the reference annual cooling demand, expressed in kWh;

H_{TO} , H_{SB} , H_{CK} , H_{OFF} = the number of hours the unit is considered to work in respectively thermostat off mode, standby mode, crankcase heater mode and off mode.

NOTE The number of hours to be used for several types of units is indicated in Annex D.

P_{TO} , P_{SB} , P_{CK} , P_{OFF} = the electricity consumption during respectively thermostat off mode, standby mode, crankcase heater mode and off mode, expressed in kW.

6.2 Calculation of the reference annual cooling demand (Q_C)

The Reference Annual Cooling demand (Q_C) is expressed in kWh and can be calculated as follows:

Full load ($P_{designc}$) multiplied with the number of equivalent cooling hours (H_{ce}).

$$Q_C = P_{designc} \times H_{ce} \quad (2)$$

NOTE For air-to-air units up to 12 kW cooling capacity, the number of equivalent cooling hours equals 350. This is based on occupancy scenarios for certain types of buildings and on a climate bin method which can be found in Annex D.

6.3 Calculation of reference $SEER_{on}$

The reference $SEER_{on}$ is determined as follows:

$$SEER_{on} = \frac{\sum_{j=1}^n h_j \cdot P_c(T_j)}{\sum_{j=1}^n h_j \cdot \left(\frac{P_c(T_j)}{EER_{PL}(T_j)} \right)} \quad (3)$$

where

T_j = the bin temperature;

j = the bin number;

n = the amount of bins;

$P_c(T_j)$ = the cooling demand of the building for the corresponding temperature T_j ;

h_j = the number of bin hours occurring at the corresponding temperature T_j ;

$EER(T_j)$ = the EER values of the unit for the corresponding temperature T_j .

The values to be used for j , T_j and h_j are determined in Table 36:

Table 36 — Bin number j, outdoor temperature Tj in °C and number of hours per bin hj corresponding to the reference cooling season

j	#	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Tj	°C	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
hj	h	205	227	225	225	216	215	218	197	178	158	137	109	88	63	39	31	24	17	13	9	4	3	1	0

The cooling demand $P_c(T_j)$ can be determined by multiplying the full load value ($P_{designc}$) with the part load ratio % for each corresponding bin. This part load ratio % is calculated as follows:

$$\text{Part load ratio \%} = (T_j - 16) / (35 - 16) \quad (4)$$

The EER values at each bin are determined via interpolation of the EER values at part load conditions A, B, C and D as mentioned in the tables of Clause 4.

For part load conditions above part load condition A, the same EER values as for condition A are used.

For part load conditions below part load condition D, the same EER values as for condition D are used.

6.4 Calculation procedure for determination of EER_{PL} values at part load conditions B, C, D

6.4.1 General

In part load condition A (full load), the declared capacity of a unit is considered equal to the cooling load ($P_{designc}$).

In part, load conditions B, C and D, there can be two possibilities:

- if the declared capacity of a unit is matching with the required cooling loads, the corresponding EER_{DC} value of the unit is to be used. This may occur with variable capacity units;
- if the declared capacity of a unit is higher than the required cooling loads, the unit has to cycle on/off. This may occur with fixed capacity or variable capacity units. In such cases, a degradation factor (C_d or C_c) has to be used to calculate the corresponding EER_{PL} value. Such calculation is explained below.

6.4.2 For air-to-air & water-to-air units

6.4.2.1 Calculation procedure for fixed capacity units

For each part load conditions B, C and D, the EER is calculated as follows:

$$EER_{PL(B,C,D)} = EER_{DC} \times (1 - C_d \times (1 - CR)) \quad (5)$$

where

EER_{DC} = the EER corresponding to the declared capacity (DC) of the unit at the same temperature conditions as for part load conditions B, C and D;

C_d = the degradation coefficient;

CR = the capacity ratio.

The capacity ratio is the ratio of the cooling demand (P_c) over the declared capacity (DC) of the unit at the same temperature conditions.

For determination of the Cd value, see 8.4.2.

If Cd is not determined by test then the default degradation coefficient Cd shall be 0,25.

6.4.2.2 Calculation procedure for variable capacity units

Determine the declared capacity and EER_{PL} at the closest step or increment of the capacity control of the unit to reach the required cooling load. If this step does not allow to reach the required cooling load within $\pm 10\%$ (e.g. between 9,9 kW and 8,1 kW for a required cooling load of 9 kW), determine the capacity and EER_{PL} at the defined part load temperatures for the steps on either side of the required cooling load. The part load capacity and the EER_{PL} at the required cooling load are then determined by linear interpolation between the results obtained from these two steps.

If the smallest control step of the unit is higher than the required cooling load, the EER_{PL} at the required part load ratio is calculated using Equation (5) as for fixed capacity units.

6.4.3 Air-to-water, water-to-water & brine-to-water units

6.4.3.1 Calculation procedure for fixed capacity units

For each part load conditions B, C and D, the EER_{PL} is calculated as follows:

$$EER_{PL(B,C,D)} = EER_{DC} \times \frac{CR}{C_c \times CR + (1 - C_c)} \quad (6)$$

where

EER_{DC} = the EER corresponding to the declared capacity (DC) of the unit at the same temperature conditions as for part load conditions B, C and D;

C_c = the degradation coefficient;

CR = the capacity ratio.

The capacity ratio is the ratio of the cooling demand (P_c) over the declared capacity (DC) of the unit at the same temperature conditions.

For determination of the C_c value, see 8.4.3.

If C_c is not determined by test then the default degradation coefficient C_c shall be 0,9.

6.4.3.2 Calculation procedure for variable capacity control units

Determine the declared capacity and EER_{PL} at the closest step or increment of the capacity control of the unit to reach the required cooling load. If this step does not allow to reach the required cooling load within $\pm 10\%$ (e.g. between 9,9 kW and 8,1 kW for a required cooling load of 9 kW), determine the capacity and EER_{PL} at the defined part load temperatures for the steps on either side of the required cooling load. The part load capacity and the EER_{PL} at the required cooling load are then determined by linear interpolation between the results obtained from these two steps.

If the smallest control step of the unit is higher than the required cooling load, the EER_{PL} at the required part load ratio is calculated using Equation (6) as for fixed capacity units.

7 Calculation methods for reference SCOP, reference SCOP_{on} and reference SCOP_{net}

7.1 General formula for calculation of reference SCOP

The reference SCOP is defined as:

Reference SCOP = reference annual heating demand divided by the annual electricity consumption.

This annual electricity consumption includes the power consumption during active mode, thermostat off mode, standby mode, off mode and that of the crankcase heater.

The power consumption during active mode is derived from the calculation of the SCOP_{on}. For determination of SCOP_{on}, see 7.3.

The calculation of the reference SCOP that applies to all types of units is given by the following formula:

$$\text{SCOP} = \frac{Q_h}{\frac{Q_h}{\text{SCOP}_{\text{on}}} + H_{\text{TO}} \times P_{\text{TO}} + H_{\text{SB}} \times P_{\text{SB}} + H_{\text{CK}} \times P_{\text{CK}} + H_{\text{OFF}} \times P_{\text{OFF}}} \quad (7)$$

where

Q_h = the reference annual heating demand, expressed in kWh;

$H_{\text{TO}}, H_{\text{SB}}, H_{\text{CK}}, H_{\text{OFF}}$ = the number of hours the unit is considered to work in respectively thermostat off mode, standby mode, crankcase heater mode and off mode;

NOTE The number of hours to be used for several types of units is indicated in Annex D.

$P_{\text{TO}}, P_{\text{SB}}, P_{\text{CK}}, P_{\text{OFF}}$ = the electricity consumption during respectively thermostat off mode, standby mode, crankcase heater mode and off mode, expressed in kW.

7.2 Calculation of the reference annual heating demand (Q_h)

The reference annual heating demand is expressed in kWh and can be calculated as follows:

Full load in heating (P_{design_h}) multiplied with the number of equivalent heating hours (H_{HE})

$$Q_h = P_{\text{design}_h} \times H_{\text{he}} \quad (8)$$

NOTE The numbers of equivalent heating hours for the average, warmer and colder reference heating seasons for certain types of units are based on occupancy scenarios for certain types of buildings and a climate bin method which can be found in Annex D.

7.3 Calculation of reference SCOP_{on} and reference SCOP_{net}

The reference SCOP_{on} and SCOP_{net} are determined as follows:

$$\text{SCOP}_{\text{on}} = \frac{\sum_{j=1}^n h_j \cdot \text{Ph}(T_j)}{\sum_{j=1}^n h_j \cdot \left(\frac{\text{Ph}(T_j) - \text{elbu}(T_j)}{\text{COP}_{\text{PL}}(T_j)} + \text{elbu}(T_j) \right)} \quad (9)$$

$$\text{SCOP}_{\text{net}} = \frac{\sum_{j=1}^n h_j \cdot (\text{Ph}(T_j) - \text{elbu}(T_j))}{\sum_{j=1}^n h_j \cdot \left(\frac{\text{Ph}(T_j) - \text{elbu}(T_j)}{\text{COP}_{\text{PL}}(T_j)} \right)} \quad (10)$$

where

T_j = the bin temperature;

j = the bin number;

n = the amount of bins;

$\text{Ph}(T_j)$ = the heating demand of the building for the corresponding temperature T_j , expressed in kW;

h_j = the number of bin hours occurring at the corresponding temperature T_j ;

$\text{COP}_{\text{PL}}(T_j)$ = the COP values of the unit for the corresponding temperature T_j ;

$\text{elbu}(T_j)$ = the required capacity of an electric backup heater for the corresponding temperature T_j , expressed in kW.

The values to be used for j , T_j and h_j are determined in Table 37:

Table 37 — Bin number j, outdoor temperature T_j in °C and number of hours per bin h_j corresponding to the reference heating seasons “warmer”, “average”, “colder”

j #	T_j °C	Warmer (W)	Average (A)	Colder (C)
		h_{jW} h	h_{jA} h	h_{jC} h
1 to 8	-30 to -23	0	0	0
9	-22	0	0	1
10	-21	0	0	6
11	-20	0	0	13
12	-19	0	0	17
13	-18	0	0	19
14	-17	0	0	26
15	-16	0	0	39
16	-15	0	0	41
17	-14	0	0	35
18	-13	0	0	52
19	-12	0	0	37
20	-11	0	0	41
21	-10	0	1	43
22	-9	0	25	54
23	-8	0	23	90
24	-7	0	24	125
25	-6	0	27	169
26	-5	0	68	195
27	-4	0	91	278
28	-3	0	89	306
29	-2	0	165	454
30	-1	0	173	385
31	0	0	240	490
32	1	0	280	533
33	2	3	320	380
34	3	22	357	228
35	4	63	356	261
36	5	63	303	279
37	6	175	330	229
38	7	162	326	269
39	8	259	348	233
40	9	360	335	230
41	10	428	315	243
42	11	430	215	191
43	12	503	169	146
44	13	444	151	150
45	14	384	105	97
46	15	294	74	61
total		3 590	4 910	6 446

The heating demand $Ph(T_j)$ can be determined by multiplying the full load value ($P_{designh}$) with the part load ratio % for each corresponding bin. This part load ratio % is calculated as follows:

- for the average climate: Part load ratio % = $(T_j - 16) / (-10 - 16)$ %;
- for the warmer climate: Part load ratio % = $(T_j - 16) / (+2 - 16)$ %;
- for the colder climate: Part load ratio % = $(T_j - 16) / (-22 - 16)$ %.

The COP_{PL} values and capacity values at each bin are determined via interpolation of the COP_{PL} and capacity values at part load conditions A, B, C and D, and in some cases also E, F. Interpolation is done between the COP_{PL} s and capacities of the 2 closest part load conditions (as mentioned in the tables of Clause 5).

The COP_{PL} values and capacity values for part load conditions above D are extrapolated from the COP_{PL} values and capacity values at part load conditions C and D.

In case of the cold climate, and if the TOL (operation limit) is below $-20\text{ }^{\circ}\text{C}$, an additional calculation point has to be taken from the capacity and COP_{PL} at $-15\text{ }^{\circ}\text{C}$ condition.

However, If the capacity of the heat pump is lower than the value of $Ph(T_j)$, correction needs to be made for the missing capacity with an electric back up heater with a COP of 1. This correction does not apply for the calculation of $SCOP_{net}$.

Below TOL (operation limit) the heat pump is not running. The capacity of the heat pump at outside air temperatures below TOL is 0 kW.

7.4 Calculation procedure for determination of COP_{PL} values at part load conditions A, B, C, D

7.4.1 General

In part, load conditions A, B, C and D, there can be two possibilities:

- if the declared capacity of a unit is matching with or lower than the required heating demands, the corresponding COP_{DC} value of the unit is to be used;
- if the declared capacity of a unit is higher than the required heating demand, COP_{PL} shall be calculated according to the calculation methods below, depending on the capacity control of the unit.

7.4.2 For air-to-air, brine-to-air and water-to-air units

7.4.2.1 Calculation procedure for fixed capacity units

For the part load conditions A, B, C, D where the capacity ratio is lower than 1, the COP_{PL} is calculated as follows:

$$COP_{PL(A,B,C,D)} = COP_{DC} \times (1 - Cd \times (1 - CR)) \quad (11)$$

where

COP_{DC} = the COP corresponding to the declared capacity (DC) of the unit at the same temperature conditions as for part load conditions A, B, C, D;

Cd = the degradation coefficient;

CR = the capacity ratio,

The capacity ratio is the ratio of the heating demand (Ph) over the declared capacity (DC) of the unit at the same temperature conditions.

For determination of the Cd value, see 8.4.2.

If Cd is not determined by test then the default degradation coefficient Cd shall be 0,25.

7.4.2.2 Calculation procedure for variable capacity units

- Determine the declared capacity and COP_{PL} at the closest step or increment of the capacity control of the unit to reach the required heat load. If this step does not allow to reach the required heating load within $\pm 10\%$ (e.g. between 9,9 kW and 8,1 kW for a required heating load of 9 kW), determine the capacity and COP_{PL} at the defined part load temperatures for the steps on either side of the required heating load. The part load capacity and the COP_{PL} at the required heating load are then determined by linear interpolation between the results obtained from these two steps.
- If the smallest control step of the unit is higher than the required heating load, the COP_{PL} at the required part load ratio is calculated using Equation (11) as for fixed capacity units.

7.4.3 For air-to-water, water-to-water and brine-to-water units

7.4.3.1 Calculation procedure for fixed capacity units

For each part load conditions A, B, C, D the COP_{PL} is calculated as follows:

$$COP_{PL(A, B, C, D)} = COP_{DC} \times \frac{CR}{C_c \times CR + (1 - C_c)} \quad (12)$$

where

COP_{DC} = the COP corresponding to the declared capacity (DC) of the unit at the same temperature conditions as for part load conditions A, B, C and D;

C_c = the degradation coefficient;

CR = the capacity ratio.

The capacity ratio is the ratio of the heating demand (Ph) over the declared capacity (DC) of the unit at the same temperature conditions.

For determination of the C_c value, see 8.4.3.

If C_c is not determined by test then the default degradation coefficient C_c shall be 0,9.

7.4.3.2 Calculation procedure for variable units

- Determine the declared capacity and COP_{PL} at the closest step or increment of the capacity control of the unit to reach the required heat load. If this step does not allow to reach the required heating load within $\pm 10\%$ (e.g. between 9,9 kW and 8,1 kW for a required heating load of 9 kW), determine the capacity and COP_{PL} at the defined part load temperatures for the steps on either side of the required heating load. The part load capacity and the COP_{PL} at the required heating load are then determined by linear interpolation between the results obtained from these two steps.
- If the smallest control step of the unit is higher than the required heating load, the COP_{PL} at the required part load ratio is calculated using Equation (12) as for fixed capacity units.

Annex D (informative)

Determination of reference annual cooling/heating demands and determination of hours for active mode, thermostat off, standby, off mode and crankcase heater mode

D.1 General track

Below information applies to reversible air-to-air units not exceeding 12 kW cooling capacity (according to EN 14511 rating condition) and heating only air-to-air units not exceeding 12 kW heating capacity.

D.2 Determination of reference annual cooling/heating demands and hours for active mode, thermostat off, standby, off mode

In general, the annual cooling/heating demand of buildings is mainly determined by:

- a) the climate : including weather data (temperatures, amount of clouds etc.) and solar radiation data
- b) the type of building and its corresponding occupancy hours, e.g. residential, retail, office etc.
- c) cooling and heating set points and set back settings
- d) the building characteristics, e.g. exposed surface, window glazing, insulation, ventilation, etc.
- e) internal gains, e.g. lighting, electrical equipment, occupancy (people/m²)

For determination of the reference annual cooling/heating demands of above-mentioned units, following assumptions are taken:

- f) climate: the reference cooling and heating seasons are represented by bins, as mentioned in Tables 29 and 30.
- g) type of building: 3 building types are used : residential, small office and small retail (shop). In general, residency occupancy hours are assumed between 7-9 am and between 5-11 pm for working days and between 9 am and 11 pm for weekends. Offices have assumed occupancy hours between 8 am and 7 pm during workdays only, whereas shops have assumed occupancy hours between 8 am and 7 pm during working and weekend days. More details on the assumed occupancy hours can be found in [2], pages 65 & 66 Tables 4-16, 4-17 and 4-18.
- h) Building characteristics: both old and new building constructions have been taken into account. More details on the assumed geometry, construction values etc can be found in [2], pages 63, 64, 65 Tables 4-11, 4-12, 4-13, 4-14.
- i) Set point and Set back settings: the cooling set point is assumed a weighted average of 25 °C and 26 °C for offices and residential dwellings (depending on EU climates) and at 23 °C for shops. The heating set point is 21 °C. Set back for heating is at 12 °C for offices and shops and at 15 °C for residences. More details can be found in [2], page 70. For the calculation of reference SEER/SCOP, setback is taken into account by reducing the number of active hours occurring during the cooling/heating season.
- j) Internal gains: more details on the impact of lighting, equipment and occupancy (people/m²) can be found in [2], page 65 Tables 4-15.

Based on the above elements, the bin hours of the cooling and heating season are determined, and also the reference numbers of hours that the units work in active mode, thermostat off mode, standby mode and off mode.

A differentiation is made between units that have only a cooling function, only a heating function or both cooling and heating functions. It is assumed that cooling only units remain connected to the mains power source outside the cooling season, and heating only units outside the heating season. In such cases, they are considered to be in "off mode". Units with both cooling and heating functions have zero "off mode" hours.

In addition, differentiation is made between the calculation of reference SEER - which is based on 1 reference climate condition only -, and the calculation of reference SCOP - which is based on 3 reference climate conditions which are different from the condition for reference SEER. As a result, the values of reference SEER and reference SCOP cannot be put together to calculate an annual seasonal performance value. See below tables for more details.

Table D.1 — Number of hours used for calculation of reference SEER

		Cooling only	Reversible
A	Total hours per year	8 760	8 760
B	Off mode (H_{off})	5 088	0
C	Difference (A-B) = hours for the reference cooling season, of which:	3 672	3 672
D	Thermostat off (H_{trO})	221	221
E	Standby(H_{SB})	2 142	2 142
F	Difference (C-D-E) = Active mode hours without setback correction	1 309	1 309
G	Setback correction	355	355
H	Difference (F-G) = (or F × 73 %) = Active mode hours corrected for setback impact	954	954
I	Equivalent active hours (H_{CE})	350	350

"Equivalent hours" means that for calculation of the reference annual cooling demand Q_{ce}:

Multiplying the 954 h of the active mode with the corresponding part load values for each bin (cf. the bins in Table 29)

is equivalent to:

multiplying 350 h with the full load value corresponding to 35 °C outdoor temperature.

A similar approach is taken for heating for calculation of reference SCOP, resulting in below values for the Average (A), Warm (W) and Cold (C) climate:

Table D.2 — Number of hours used for calculation of reference SCOP

	Heating only			Reversible		
	"A"	"W"	"C"	"A"	"W"	"C"
Off mode (Hoff)	3 672 h	4 345 h	2 189 h	0 h	0 h	0 h
Thermostat off (Hto)	179 h	755 h	131 h	179 h	755 h	131 h
Standby(Hsb)	0 h	0 h	0 h	0 h	0 h	0 h
Equivalent active hours for heating (Hhe)	1 400 h	1 400 h	2 100 h	1 400 h	1 400 h	2 100 h

D.3 Hours used for crankcase heater mode

The number of hours for the crankcase heater mode for above-mentioned units is as below:

Table D.3 — Crankcase heater mode hours for determination of reference SEER

	Cooling only	Reversible
Crankcase heater (Hck)	7760	2672

Table D.4 — Crankcase heater mode hours for determination of reference SCOP

	Heating only			Reversible		
	"A"	"W"	"C"	"A"	"W"	"C"
Crankcase heater (Hck)	3 851 h	4 476 h	2 944 h	179 h	755 h	131 h